Intermittency in superfluid helium counterflow turbulence

P. Z. Stasiak
School of Mathematics, Statistics and Physics, Newcastle University, Newcastle upon Tyne, NE1 7RU, United Kingdom

G. Krstulovic
Université Côte d'Azur, Observatoire de la Côte d'Azur, CNRS,Laboratoire Lagrangre, Boulevard de l'Observatoire CS 34229 - F 06304 NICE Cedex 4, France

L. Galantucci
Istituto per le Applicazioni del Calcolo "M. Picone" IAC CNR, Via dei Taurini 19, 00185 Roma, Italy

(Dated: June 23, 2026)

INTRODUCTION

The chaotic dynamics of turbulent fluids is a ubiquitous phenomenon dominating a vast array of physical systems in nature [1]. Turbulence shapes the patterns and properties of fluids of all kinds, from ordinary viscous fluids in Navier-Stokes turbulence [2] to electrically conducting fluids in magneto-hydrodynamics [3] and quantum fluids in quantum turbulence [4, 5]. Such systems are characterised by the existence of a wide range of length scales across which inviscid conserved quantities, such as energy, are transferred without any loss in the spirit of the cascade described by Richardson [6]. In 3D classical turbulence this process is a direct cascade where energy is transferred from large to small scale structures at a constant rate of energy with a well-known scaling law of Kolmogorov's phenomenological K41 theory [7]. The dynamics of these filament structures can result in collisions known as vortex reconnections, altering the topology of the system and re-distributing energy between scales and fluid components [8, 9]. The richness of turbulence in quantum fluids arises from the additional length scale ℓ, separating the dynamics at large scales where the flow is unaffected by small-scale vortex structures, and small scales where the flow is strongly dependent on them. At these large scales, the same cascade phenomenology of the K41 theory has been observed in quantum fluids [10], and the turbulent flows have been known to behave intermittently [5]. Velocity increments $\delta\mathbf{v}(\mathbf{r}) = \mathbf{v}(\mathbf{x}+\mathbf{r})-\mathbf{v}(\mathbf{x})$ and corresponding structure functions $S_p = \langle|\delta\mathbf{v}(\mathbf{r}^p|\rangle$ recover scaling exponents $S_p \sim r^{\zeta_p}$ in the inertial range with striking similarity to classical turbulence [11, 12].

Recent work in classical turbulence [13] and quantum turbulence [14–17] have demonstrated the success of velocity circulation Γ given by

$$\Gamma_r(\mathbf{v}) = \oint_{\mathcal{C}(r)} \mathbf{v}\cdot d\mathbf{r} = \oiint_{\mathcal{A}(r)} (\nabla\times\mathbf{v})\cdot\hat{\mathbf{n}}dA, \qquad (1)$$

where $\mathcal{C}(r)$ denotes a closed loop enclosing an around area $\mathcal{A}(r)$, as an alternative to velocity increments for the analysis of intermittency. For higher order moments, high-Reynolds number simulations have demonstrated that circulation scaling exponents are well described by monofractal fits [13], exhibiting a simpler statistical structure in the inertial range than the multi-fractal scaling of velocity increments. In fact, Gross-Pitaevskii (GP) simulations [14, 15] and finite-temperature grid turbulence in ^{4}He [16] display a scaling of circulation variance $\langle|\Gamma_r|^2\rangle \sim r^{8/3}$ within the inertial range, coinciding with classical turbulence [1].

In this Letter, we compute the velocity circulation statistics for steady-state counterflow turbulence in ^{4}He and compare our findings with a purely superfluid Taylor-Green simulation. Counterflow turbulence differs significantly from mechanically generated turbulence in two key aspects. Firstly, the imposed counterflow velocity facilitates an energy injection mechanism by the Donnelly-Glaberson instability [8, 18]. Secondly, steady-state counterflow turbulence is known to be strongly anisotropic, and has a tendency to develop large-scale quasi-two-dimensional structures [15, 19] which can support an inverse cascade of energy [20]. To account for this anisotropy, we split our analysis of the counterflow state into the orientations parallel and perpendicular to the counterflow direction.

METHOD

Superfluid dynamics are modelled by employing the recently developed FOUCAULT model [21]. We follow Schwarz's approach [22], by exploiting the large separation of scales between the ^{4}He vortex core size a_0 and the average inter-vortex spacing ℓ. We parametrize the superfluid vortex lines as 1D space curves $\mathbf{s}(\xi,t)$ with ξ and t the arclength and time respectively. The corresponding equation of motion is

$$\dot{\mathbf{s}}(\xi,t) = \mathbf{v}_s + \frac{\rho_n}{\rho}\left[\mathbf{v}_{ns}\cdot\mathbf{s}'\right]\mathbf{s}' + \beta\mathbf{s}'\times\mathbf{v}_{ns} + \beta'\mathbf{s}'\times\left[\mathbf{s}'\times\mathbf{v}_{ns}\right] \qquad (2)$$

where $\mathbf{s}' = \partial\mathbf{s}/\partial\xi$ is the unit tangent vector at $\mathbf{s}$, $\mathbf{v}_{ns} = \mathbf{v}_n - \mathbf{v}_s$ where $\mathbf{v}_n$ and $\mathbf{v}_s$ are the normal fluid

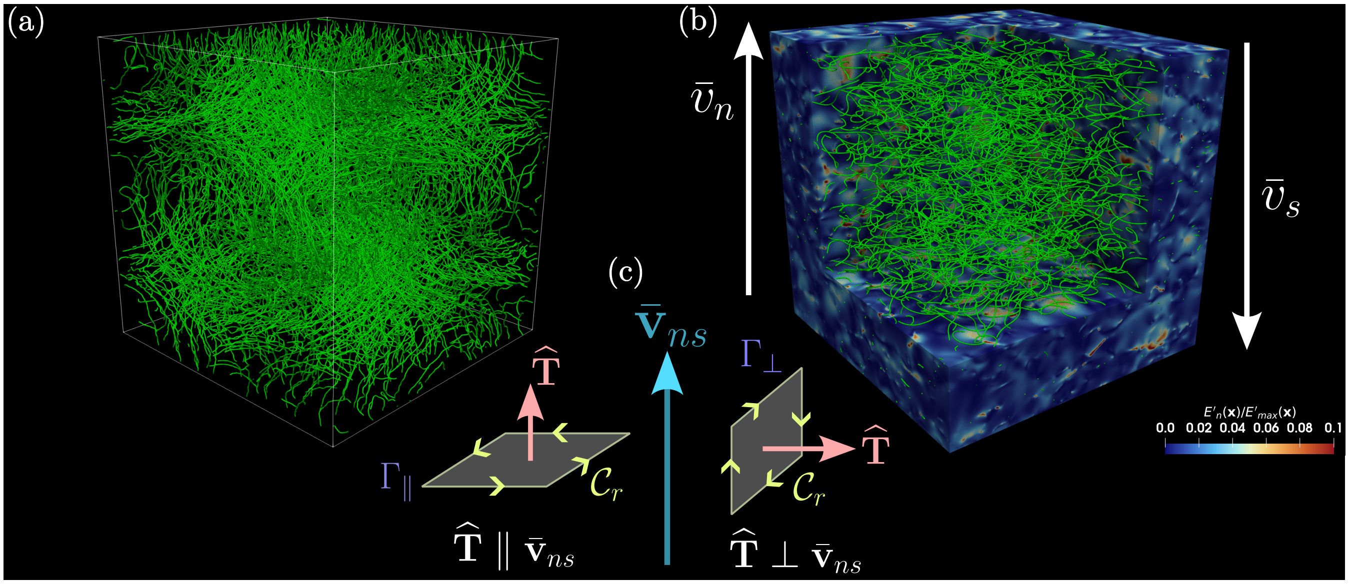

FIG. 1: Visualisation of turbulent vortex tangles. (a) Counterflow-induced turbulence generated at $T = 2.1$ K with a counterflow velocity of $\bar{v}_{ns} = \bar{v}_n - \bar{v}_s = 0.94$ cm/s. The blue and red surface rendering shows the normalised kinetic energy density of normal fluid fluctuations E'_n/E'_{max} where $E'_n = |\mathbf{v}_n - \bar{\mathbf{v}}_n|^2$. (b) Turbulence generated by an initial Taylor-Green configuration (see Supplementary Materials for details) $T = 0$K. In both (a) and (b) superfluid vortex lines are represented as green tubes with exaggerated size. (c) Schematic diagram of the parallel and perpendicular slices when computing the circulation. Given a closed square path $\mathcal{C}_r$ of length r and a unit normal to the enclosed surface $\widehat{\mathbf{T}}$, circulation statistics are averaged based on the orientation of $\widehat{\mathbf{T}}$ with respect to the mean counterflow velocity $\bar{\mathbf{v}}_{ns}$.

and superfluid velocities at $\mathbf{s}$, and β, β' are temperature and Reynolds number dependent mutual friction coefficients. The superfluid velocity $\mathbf{v}_s$ is computed via a desingularized Biot-Savart integral, accelerated using the tree algorithm [23] (see Supplementary Materials). We describe the normal fluid in a classical way, using the incompressible ($\nabla \cdot \mathbf{v}_n = 0$) Navier-Stokes equation

$$\frac{\partial \mathbf{v}_n}{\partial t} + (\mathbf{v}_n \cdot \nabla)\mathbf{v}_n = -\frac{1}{\rho}\nabla p + \nu_n \nabla^2 \mathbf{v}_n + \frac{\mathbf{F}_{ns}}{\rho_n} \qquad (3)$$

where ρ_n and ρ_s are the normal fluid and superfluid densities, $\rho = \rho_n + \rho_s$, p is the pressure, ν_n is the kinematic viscosity of the normal fluid and $\mathbf{F}_{ns}$ is the mutual friction force per unit volume.

In this study, we prepare two distinct turbulent vortex tangles generated by (a) a $T = 2.1$ K thermal counterflow (CF) with an average inter-vortex distance ℓ_{CF} and (b) a $T = 0$ K evolution of a tangle with a Taylor-Green initial condition (TG) with an average inter-vortex spacing ℓ_{TG}. Both of these tangles are visualized in the left and right panels of Fig. 1 respectively. The velocity circulation statistics of $\mathbf{v}_s$ are obtained by a vorticity coarse-graining process described in the Supplementary materials. Furthermore, the statistics related to the counterflow are decomposed into the normal fluid (NF) and superfluid (SF) components. Each of these components are further decomposed into the perpendicular ($\perp$) and ($\parallel$) components, corresponding to the directions over which statistics are averaged. Perpendicular (parallel) relates to the statistics averaged in the direction where the flux of vorticity through a square loop of size r is perpendicular (parallel) to the counterflow velocity $\mathbf{v}_{ns}$. The circulation statistics are computed using the Circulation.jl package [**?**].

RESULTS

Firstly, we begin by considering the simplest observable for velocity circulation statistic - the variance $\langle|\Gamma|^2\rangle$ for different loop sizes r. The circulation variance scalings are shown in Fig. 2, where we have identified two scaling regimes, a quantum regime $r \ll \ell$ with red shading and a classical regime $r \gtrsim \ell$ with blue shading. Indeed in the quantum regime we find good agreement for the superfluid velocity statistics with the small-scale theory $\langle|\Gamma^2|\rangle \sim r^2$ as discussed in Ref. [14] using the Gross-Pitaevskii (GP) model. We observe that in this regime the scaling not only follows the GP theory, but the quantum region scaling is independent of thermal effects and the nature of the turbulent driving force. Similarily, the normal fluid also exhibits the well known viscous scaling of $\langle|\Gamma|^2|\rangle \sim r^4$ for small loop sizes r , as a result of smoothness at very small scales [13].

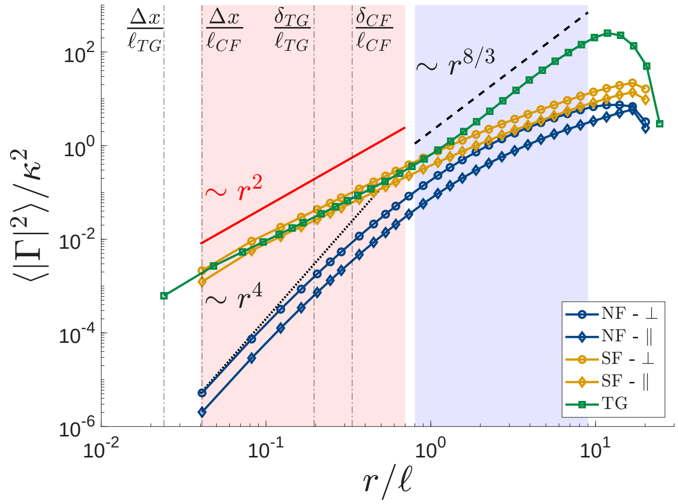

FIG. 2: Circulation variance $\langle|\Gamma|^2\rangle$ around square loops of size r. The blue and yellow curves show the normal fluid and super fluid simulations respectively, which are subdivided into the perpendicular (circles) and parallel (diamonds) components. The green squares show the Taylor-Green simulation.

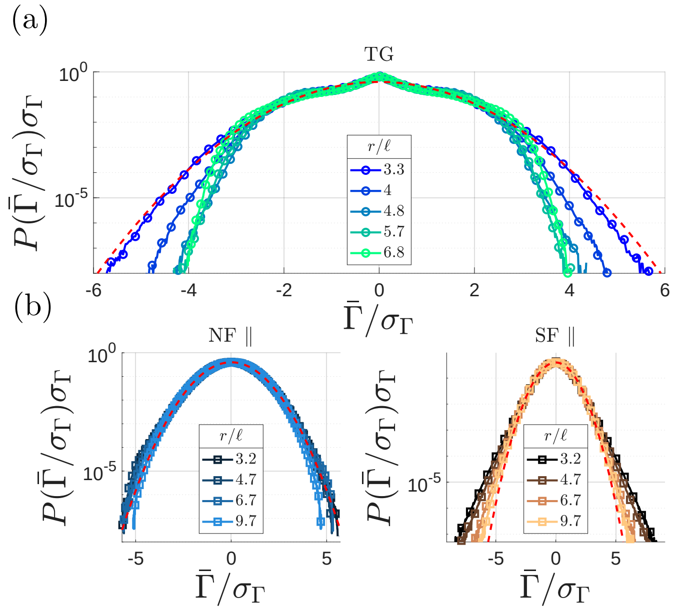

FIG. 3: PDFs of the velocity circulation for different loop sizes $r/\ell > 1$, normalized by the standard deviation σ_Γ. (a) shows the Taylor-Green simulation, while (b) shows the parallel components of the normal fluid and superfluid respectively.

The cross-over from the quantum $r \ll \ell$ to the classical $r \gtrsim \ell$ marks a stark contrast in the superfluid circulation variance scaling. While the TG scales in accordance with a partial polarization of vortices where $r^{8/3}$ [24], the CF scaling, appears to follow a weaker scaling than a fully random tangle of vortex lines that exhibits a r^2 scaling in the classical range. Remarkably both components of the SF and NF statistics of CF appear to adhere to this scaling law for loops $r \gtrsim \ell$. For the remainder of this Letter, we will only focus on the classical range where the loop sizes $r > \ell$.

In fully-developed isotropic classical turbulence, velocity circulation statistics adhere to a strictly non-Gaussian distribution with long tails [25]. Deviations from Gaussian statistics are the hallmark of intermittency, and represent a deviation from self-similar scaling which are evident in the TG simulations shown in Fig. 3a. The circulation PDF of the NF components differs very strongly (see Fig. 3b). Strikingly, we see a near collapse to Gaussian statistics for the normal fluid component, while the parallel component exhibits very small deviations. ~~We show this collapse by computing the flatness $\mathcal{K}(\Gamma) = \langle|\Gamma|^4\rangle/\sigma_\Gamma^4$, where σ_Γ is the standard deviation $\sqrt{\langle|\Gamma|^2\rangle}$. The flatness is displayed in the inset of Fig. 3a. In the classical regime the NF components monotonically decay to the Gaussian value of $\mathcal{K} = 3$, while the non-Gaussian TG simulation deviates in the same region. Indeed we observe that the presence of quantized vortices suppresses the intermittent behaviour of the classically-driven NF.~~

As is customary when the scaling is not clear, we will characterize the intermittency by measuring the deviation of the circulation moments of order p given by $\langle|\Gamma|^p\rangle$ from the self-similar K41 scaling. The TG and NF moments are shown in Fig. 4, where we have plotted the circulation moments in extended self-similarity (ESS) using the third-order circulation moment $\langle|\Gamma|^2\rangle$. In this way, we define the scaling exponents λ_p as

$$\langle|\Gamma|^p\rangle \sim \left[\langle|\Gamma|^2\rangle\right]^{(\lambda_p/\lambda_2)p} \tag{4}$$

In order to compute the scaling exponents, we find the local slope of the moments in ESS, which is given by

$$\left(\frac{\lambda_p}{\lambda_2}\right)(r) = \frac{\mathrm{d}\left[\log\langle|\Gamma|^p\rangle\right]}{\mathrm{d}\left[\log\langle|\Gamma|^2\rangle\right]} \tag{5}$$

As the local-slope of pure power-law functions is flat, we look for flat regions where the loop sizes are $r > \ell$, and then average over the selected regions. The scaling exponents λ_p are shown in Fig. 5 for integer moments of order p. As expected, the TG scaling exponents lie closely to the bifractal behaviour identified in high Reynolds number classical turbulence [13] and also in pure GPE quantum turbulence [14]. The bifractal fit for classical turbulence intermittency uses a robust linear scaling $\lambda_p = \alpha p$ where $\alpha \approx 1.367$ for low order moments $p < 3$, which is in good agreement with our findings. For larger moments, we see that both the TG and VF circulation moments can be closely related to the monofractal fit $\lambda_p = hp + (3-D)$ with fractal dimension D and Hölder exponent h with weak Reynolds number dependence. For the bifractal fit

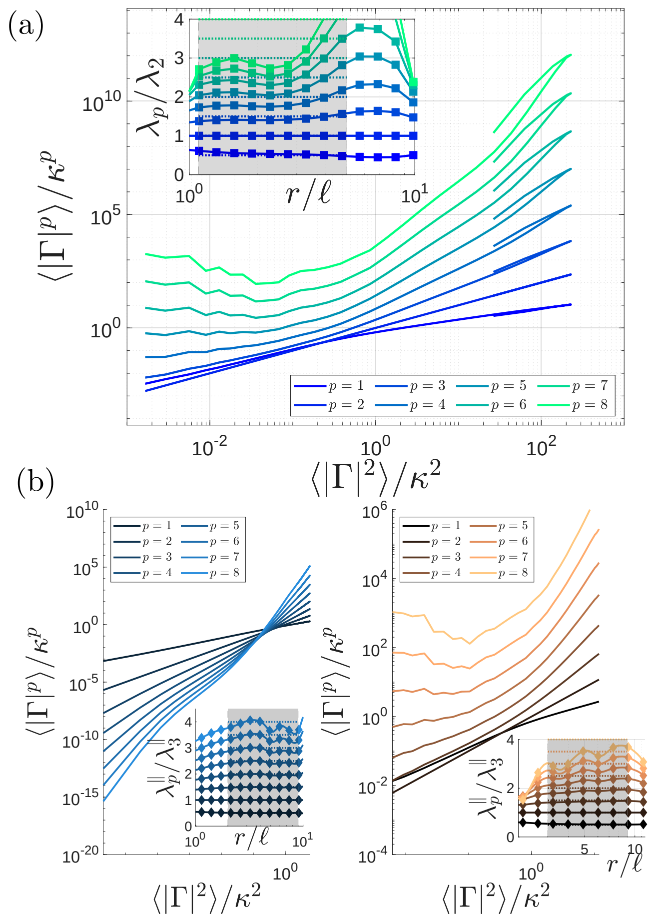

FIG. 4: Circulation moments $\langle|\Gamma|^p\rangle$ in extended self-similarity (ESS) using $p=2$. (a) shows the Taylor-Green simulation, while the left and right panels of (b) are the normal fluid perpendicular and parallel component respectively. The insets of each plot give the local slope, where $\lambda_p/\lambda_3 = \mathrm{d}\left[\log\langle|\Gamma|^p\rangle\right]/\mathrm{d}\left[\log\langle|\Gamma|^3\rangle\right]$ and the horizontal lines correspond to the K41 scalings $\lambda_p^{K41} = 4p/3$. The shaded regions indicate the classical region where the data was used to estimate the exponents.

used in Fig. 5, we used $D \approx 2.2$ and $h \approx 1.1$, which correspond to the values estimated in the highest Reynolds number flow in Ref. [13].

We make two major remarks about CF scaling exponents. Firstly, we can see a striking departure between the NF and VF statistics. While the VF components closely follow the intermittent monofractal scaling for larger moments, the NF components exhibit a collapse to self-similar K41 theory. Secondly, in both the NF and VF statistics, the parallel component shows a higher-level of intermittency than the perpendicular component. Indeed in the inset of Fig. 5, we can see from the proportional deviation from K41 theory that the NF perpendicular component almost perfectly converges to self-similarity,

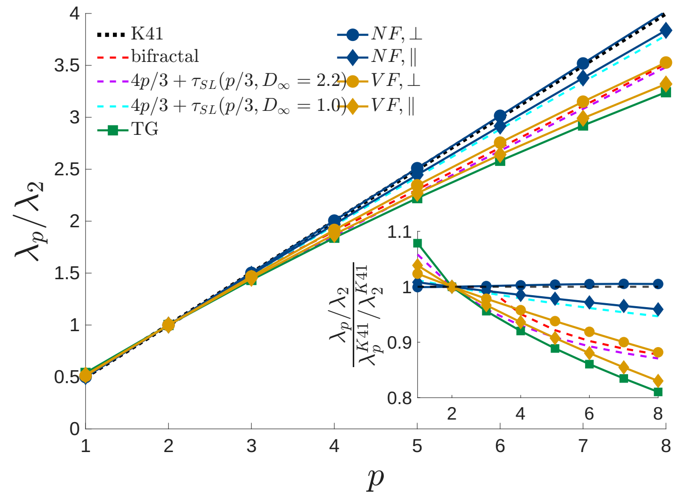

FIG. 5: Scaling exponents of velocity circulation moments, estimated within the classical range $r > \ell$. The colour scheme and linestyles is the same as in Fig. 2. The black dashed line represents the self-similar K41 scaling $\lambda_p^{K41} = 4p/3$. The inset shows the proportional deviation from K41 theory.

while the parallel component exhibits a small level of intermittency. Within ESS, a linear monofractal of the form above is under-determined, and therefore we can only estimate the ratio $a = h/(3-D)$, which for the NF parallel component gives $a \approx 9.34$. In order to describe the velocity circulation intermittency which we observe in the NF parallel component we adopt the She-Leveque log-Poisson model [26], such that the scaling exponents are given by $\lambda_p = \alpha p + C\left[1-\beta^{2/3}\right]$ where $C = 3-D$ is the codimension of the most singular structures and β characterizes the intermittency.

[1] U. Frisch, Turbulence: The Legacy of A. N. Kolmogorov (Cambridge University Press).

[2] S. Pope, Turbulent Flows (Cambridge University Press).

[3] V. M. Canuto and J. Christensen-Dalsgaard, Turbulence in astrophysics: Stars, Annu. Rev. Fluid Mech. **30**, 167 (1998).

[4] CF. Barenghi, HAJ. Middleton-Spencer, L. Galantucci, and NG. Parker, Types of quantum turbulence, AVS Quantum Sci. **5**, 025601 (2023).

[5] C. F. Barenghi, L. Skrbek, and K. R. Sreenivasan, Quantum Turbulence (Cambridge University Press, Cambridge, 2023).

[6] L. F. Richardson, Weather Prediction by Numerical Process (Cam Univ Pr, 1922).

[7] A. Kolmogorov, The Local Structure of Turbulence in an Incompressible Viscous Fluid for Very Large Reynolds Numbers, Dokl. Akad. Nauk. SSSR **30**, 301 (1941).

[8] P. Z. Stasiak, A. W. Baggaley, G. Krstulovic, C. F. Barenghi, and L. Galantucci, Cross-Component Energy

Transfer in Superfluid Helium-4, J. Low Temp. Phys. (2024).

[9] P. Z. Stasiak, Y. Xing, Y. Alihosseini, C. F. Barenghi, A. Baggaley, W. Guo, L. Galantucci, and G. Krstulovic, Experimental and theoretical evidence of universality in superfluid vortex reconnections, Proc. Natl. Acad. Sci. **122**, e2426064122 (2025).

[10] CF. Barenghi, L. Skrbek, and KP. Sreenivasan, Introduction to Quantum Turbulence, Proc. Natl. Acad. Sci. USA **111**, supp 1, 4647 (2014).

[11] J. Maurer and P. Tabeling, Local investigation of superfluid turbulence, EPL **43**, 29 (1998).

[12] E. Rusaouen, B. Chabaud, J. Salort, and P.-E. Roche, Intermittency of quantum turbulence with superfluid fractions from 0% to 96%, Physics of Fluids **29**, 105108 (2017).

[13] K. P. Iyer, K. R. Sreenivasan, and P. K. Yeung, Circulation in High Reynolds Number Isotropic Turbulence is a Bifractal, **9**, 041006.

[14] N. P. Müller, J. I. Polanco, and G. Krstulovic, Intermittency of Velocity Circulation in Quantum Turbulence, **11**, 011053.

[15] J. I. Polanco, N. P. Müller, and G. Krstulovic, Vortex clustering, polarisation and circulation intermittency in classical and quantum turbulence, **12**, 7090.

[16] N. P. Müller, Y. Tang, W. Guo, and G. Krstulovic, Velocity circulation intermittency in finite-temperature turbulent superfluid helium, Phys. Rev. Fluids **7**, 104604 (2022).

[17] N. P. Müller and G. Krstulovic, Exploring the Equivalence between Two-Dimensional Classical and Quantum Turbulence through Velocity Circulation Statistics, Phys. Rev. Lett. **132**, 094002 (2024).

[18] W. I. Glaberson, W. W. Johnson, and R. M. Ostermeier, Instability of a vortex array in He II, Phys. Rev. Lett. **33**, 1197 (1974).

[19] L. Biferale, D. Khomenko, V. L'vov, A. Pomyalov, I. Procaccia, and G. Sahoo, Superfluid Helium in Three-Dimensional Counterflow Differs Strongly from Classical Flows: Anisotropy on Small Scales, Phys. Rev. Lett. **122**, 144501 (2019).

[20] J. I. Polanco and G. Krstulovic, Counterflow-Induced Inverse Energy Cascade in Three-Dimensional Superfluid Turbulence, Phys. Rev. Lett. **125**, 254504 (2020).

[21] L. Galantucci, A. W. Baggaley, C. F. Barenghi, and G. Krstulovic, A new self-consistent approach of quantum turbulence in superfluid helium, **135**, 547.

[22] K. W. Schwarz, Three-dimensional vortex dynamics in superfluid 4He, Phys. Rev. B **38**, 2398 (1988).

[23] A. W. Baggaley and C. F. Barenghi, Tree Method for Quantum Vortex Dynamics, **166**, 3.

[24] V. S. L'vov, S. V. Nazarenko, and O. Rudenko, Bottleneck crossover between classical and quantum superfluid turbulence, **76**, 024520.

[25] M. Wilczek and R. Friedrich, Dynamical origins for non-Gaussian vorticity distributions in turbulent flows, **80**, 016316.

[26] Z.-S. She and E. Leveque, Universal scaling laws in fully developed turbulence, **72**, 336.